

Limitations of Self-Reported Admissions Data

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The Grad Cafe has important limitations for statistical analysis because it is a voluntary collection of anonymous submissions, not a random sample of graduate applicants. The submissions on this site have to come from those that know about the site, participate in online admissions discussion, and choose to disclose their outcomes. These people may differ systematically from those who never post. This makes the number of reported outcomes especially easy to misinterpret. For example, my analysis found 528 reported rejections from Stanford and 451 from Yale, etc. These are the two largest rejection counts in the database, but they are reported counts, not actual rejection rates. Stanford appearing first does not establish that it rejects a larger percentage of applications than the other universities. Instead, it could reflect on how many Grad Cafe users apply to Stanford, which applicants chose to report a rejection, or which programs and applicant populations are most active on the site. One person may also report applications to multiple universities. Consequently, the database can tell me how many rejection records it contains for each university, but it cannot tell me the universities' actual rejection rates without knowing the total and representative population of applicants.

Selection and self-report also affect statistics calculated from GPA and GRE scores. My database has an overall average GRE quantitative score of 165.79, but this average includes only the values actually reported and retained in the data. This is notably higher than the mean Quantitative score of approximately 155 among recent GRE test takers reported by Kaplan test Prep (2024). The difference does not mean that graduate applicants actually average 165.79. Instead, applicants with particularly strong scores may be more likely to disclose them on Grad Cafe, while applicants with lower scores or applicants to test-optional programs may report no GRE score at all. The issue is also visible when scores are grouped by program: finance averaged 168.39 based only on 23 reported scores, while economics averaged 167.91 based on 400. Missing values, inconsistent scoring formats, and unreliable or joke submissions create additional uncertainty because anonymous entries cannot be independently verified. Cleaning invalid values prevents some records from distorting the calculations, but it cannot make the remaining sample representative. Therefore, these statistics accurately describe the cleaned Grad Cafe records in my database, but they do not provide enough evidence to assume that the same values or patterns apply to the overall population of graduate applicants.

Kaplan Test Prep. (2024, May 16). *What's a good GRE score?*
<https://www.kaptest.com/study/gre/whats-a-good-gre-score/>